

§ Lecture 13.0

Monday, 28 September 2026

19:14

Lemma: For every real number z , there is an integer m such that

$$m-1 \leq z \leq m.$$

Proof: Since $|z| \geq 0 \Rightarrow |z|+1 > 0$.

From Archimedean property there is a positive integer M

such that $M \cdot 1 > |z|+1 > |z|$

$$\Rightarrow -M < z < M$$

So we are able to found a big range. Let us narrow

it down. We keep on adding to $-M$ and

Define

$$P(n) = \text{' if } z < -M+n \text{ then } \exists m \text{ s.t. } m-1 \leq z < m \text{'}$$

Base case $P(0)$: if $z < -M$ then $\exists m$ s.t. $m-1 \leq z < m$.

This is false, thus $m-1 \leq z < m$ is vacuously true.

Inductive part: Assume that if $z < -M+n$ then $\exists m$ s.t. $m-1 \leq z < m$.

Now for $-M+n+1$: Assume $z < -M+n+1$, then

① if $z < -M+n$ (then from hypotheses, $\exists m$ s.t. $m-1 \leq z < m$)

② if $z \geq -M+n$, then $-M+n \leq z < -M+n+1$

$$\text{take } m = -M+n+1 \Rightarrow -M+n = m-1$$

$$\Rightarrow m-1 \leq z < m.$$

$\Rightarrow P(n+1)$ is true.

Thus, the property $P(n)$ is true for all n .

Take $n=2M$

$\Rightarrow$ if $z < M$, which is true, $\exists$ integer m s.t. $m-1 \leq z < m$.

Lemma (density of rationals): Let x, y be two real numbers s.t.

$x < y$. Then there exists a rational q s.t. $x < q < y$.

Proof: Since $x < y \Rightarrow y-x > 0$.

Then from Archimedean property there exists a positive

integer N s.t.

$$N(y-x) > 1$$

$$\text{or } Ny - Nx > 1 \rightarrow \textcircled{1}$$

Since Nx is a real number, there exists integer M s.t.

$$M-1 \leq Nx < M \Rightarrow M \leq Nx+1 \text{ \& } M > Nx$$

$$\text{or } Nx < M \leq Nx+1 < Nx + Ny - Nx$$

$$\text{or } Nx < M < Ny$$

$$\Rightarrow x < \frac{M}{N} < y$$

Thus $q = \frac{M}{N}$ s.t. $x < q < y$.

Definition (Upper bound and least upper bound). Let E be a subset of $\mathbb{R}$ and M be any real number. We say that

- (i) M is an upper bound on E iff $x \leq M \quad \forall x \in E$.
- (ii) M is a least upper bound on E iff (a) M is an upper bound on E .
- (b) Any other upper bound M' on E , we have $M' \geq M$.

[Unique: $M_1 \geq M_2$ & $M_2 \geq M_1$ if both are least upper bounds]

Let us consider few examples.

① $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$.

Let M be an upper bound on $\mathbb{R}^+$. Then $x \leq M \quad \forall x \in \mathbb{R}^+$. In particular, M is positive as $1 \in \mathbb{R}^+$ and $1 \leq M$. Since M is positive $M+1 \in \mathbb{R}^+ \Rightarrow M+1 \leq M$. [contradiction]

② $E = \phi$, Every real number is an upper bound on ϕ vacuously.

But E doesn't have a least upper bound.

Let M be the least upper bound on ϕ . Since all real numbers are upper bound on ϕ , we must have $M \leq M'$ for all real M' . Choose $M' = M-1 \Rightarrow M \leq M-1$ [Contradiction]

③ $E = \{q \in \mathbb{Q} : q^2 < 2\}$.

Let $r > 0$ be a rational such that $r^2 > 2$. Assume there is $q \in E$ s.t.

$$q > r > 0 \Rightarrow q^2 = q \cdot q > r^2 > 2 \quad \text{[Contradiction]}$$

Thus any positive rational r with $r^2 > 2$ is a valid upper bound.

Now given any upper bound r on E , we can construct

$$r' = \frac{1}{2} \left(r + \frac{2}{r} \right) \quad (\text{Since } r \neq 0 \text{ as } r^2 > 2)$$

$$r'^2 = \frac{1}{4} \left[r^2 + \frac{4}{r^2} + 4 \right]$$

$$\Rightarrow r'^2 - 2 = \frac{1}{4} \left[r^2 + \frac{4}{r^2} - 4 \right]$$

$$= \frac{1}{4} \left(r - \frac{2}{r} \right)^2 > 0. \quad (\text{Since } r^2 \neq 2)$$

$\Rightarrow r'$ is also a valid upper bound.

$$\text{Now } r' - r = \frac{1}{2} \left(r + \frac{2}{r} \right) - r$$

$$= \frac{1}{2} - \frac{r}{2} = \frac{2-r^2}{2r} \quad (r^2 > 2 \Rightarrow 2-r^2 < 0) \Rightarrow r > 0$$

$$< 0 \Rightarrow r' < r.$$

Since $1 \in E$, this means all upper bounds must be positive.

Since we can construct a smaller upper bound for every upper bound,

there is no least upper bound on E in $\mathbb{Q}$.

Claim: Let E be non empty subset of $\mathbb{R}$ and $n \geq 1$ be an integer. Let $L < K$

be integers. Suppose K/n is an upper bound on E while L/n is not. Then

there exists an integer m s.t. $L < m \leq K$ s.t. $\frac{m}{n}$ is U.B but

$\frac{m-1}{n}$ is not. and m is unique.

Proof:



$\mathbb{Q}$. Is it an upper bound on some set? No No No YES YES ...

Define

$P(d) = "$ if $\frac{L+d}{n}$ is an upper bound on E , $\exists$ integer m with $L < m \leq L+d$ such that $\frac{m}{n}$ is an U.B while $\frac{m-1}{n}$ is not."

$P(0)$: if $\frac{L}{n}$ is upper bound ... Vacuously true.

Assume $P(d)$: if $\frac{L+d}{n}$ is an U.B on E , $\exists m$ s.t. $L < m \leq L+d$ s.t. $\frac{m}{n}$ is an U.B but $\frac{m-1}{n}$ is not.

$P(d+1)$: let $\frac{L+d+1}{n}$ be an U.B on E .

Case: if $\frac{L+d}{n}$ is an upper bound on $E \Rightarrow$ Claim is true.

Case: if $\frac{L+d}{n}$ is not an upper bound on E , then $m = L+d+1$ is

such that $\frac{m}{n}$ is an upper bound while $\frac{m-1}{n}$ is not.

Choose $d = K-L > 0$ natural number. Then indeed $\frac{L+K-L}{n} = \frac{K}{n}$ is an

upper bound and $\exists m$ s.t. $L < m \leq K$ s.t. $\frac{m}{n}$ is an U.B but $\frac{m-1}{n}$

is not.

Uniqueness: Let $L < m \leq K$, $L < m' \leq K$, where

$\frac{m}{n}, \frac{m'}{n}, \frac{K}{n}$ are upper bounds and $\frac{L}{n}, \frac{m-1}{n}, \frac{m'-1}{n}$ are not.

$$\text{Assume } m < m' \Rightarrow m+1 \leq m' \text{ or } \frac{m}{n} \leq \frac{m'-1}{n}$$

Since $\frac{m}{n}$ is an U.B $\Rightarrow \frac{m-1}{n}$ is U.B too [contradiction]

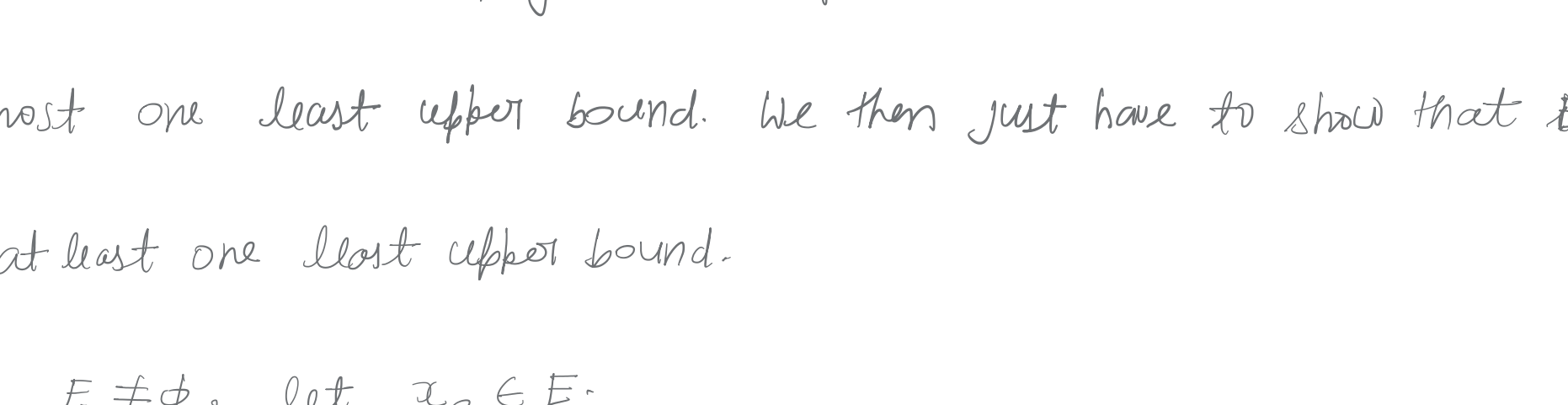
Same for $m' < m$.

$$\Rightarrow m' = m.$$

Theorem (Existence of least upper bound). Let E be a nonempty set of

$\mathbb{R}$. If E has an upper bound, then it must exactly have one least

upper bound.



Proof: Let E be a nonempty subset of $\mathbb{R}$. We know that E can have

at most one least upper bound. We then just have to show that E

has at least one least upper bound.

Since $E \neq \phi$, let $x_0 \in E$.

Let $n \geq 1$ be an integer. We know that E has an upper bound. Call

it M . Then nM is a real number. Then there exists positive integer K

$$\text{s.t. } -K < nM < K \Rightarrow \frac{K}{n} > M.$$

Since M is an upper bound on $E \Rightarrow K/n$ is also an upper bound on E .

Now since $n x_0$ is a real number there exists positive integer L' s.t.

$$-L' < n x_0 < L' \Rightarrow -\frac{L'}{n} < x_0$$

Since $x_0 \in E \Rightarrow -\frac{L'}{n}$ is not an upper bound on E .

$$\text{Call } L = -L'$$

$$\Rightarrow \frac{L}{n} < x_0 \leq M < \frac{K}{n} \Rightarrow L < K$$

Then from previous lemma there exists a unique integer m_n with

$L < m_n \leq K$ such that $\frac{m_n}{n}$ is an upper bound while $\frac{m_n-1}{n}$ is not.

Story: We get a sequence $(m_n)_{n \geq 1}$ with property that $\frac{m_n}{n}$ is upper bound

$\frac{m_n-1}{n}$ is not. for all $n \geq 1$.

Fix some $N \geq 1$. Let $n, n' \geq N$, then $\frac{m_n}{n}$ is an U.B. while $\frac{m_{n'}-1}{n'}$ is not.

$$\Rightarrow \frac{m_{n'}-1}{n'} < \frac{m_n}{n} \quad [\text{otherwise } \frac{m_{n'}-1}{n'} \geq \frac{m_n}{n} \text{ would imply } \frac{m_{n'}-1}{n'} \text{ is an U.B}]$$

$$\Rightarrow \frac{m_n}{n} - \frac{m_{n'}}{n'} > -\frac{1}{n'} \geq -\frac{1}{N}$$

$$\text{Similarly } \frac{m_{n'}}{n'} - \frac{m_n}{n} > -\frac{1}{n} \geq -\frac{1}{N}$$

$$\text{or } \frac{m_n}{n} - \frac{m_{n'}}{n'} \leq \frac{1}{N}$$

$$\Rightarrow \left| \frac{m_n}{n} - \frac{m_{n'}}{n'} \right| \leq \frac{1}{N} \quad \forall n, n' \geq N.$$

Let $\epsilon > 0$ be arbitrary. Then $\exists N$ s.t. $N\epsilon > 1 \Rightarrow \frac{1}{N} < \epsilon$.

$\Rightarrow \left(\frac{m_n}{n} \right)_{n \geq 1}$ is a Cauchy sequence.

$S = \lim_{n \rightarrow \infty} \frac{m_n}{n}$ is a real number.

$$\text{Also } S = \lim_{n \rightarrow \infty} \frac{m_n-1}{n}.$$

We need to prove that S is the least upper bound.

S is an upper bound: $\frac{m_n}{n}$ is an upper bound.

$$\Rightarrow x \leq \frac{m_n}{n} \quad \forall x$$

$$x \leq \lim_{n \rightarrow \infty} \frac{m_n}{n} = S.$$

S is least U.B: Let P is any upper bound. Since $\frac{m_n-1}{n}$ is not an U.B.

$$P \geq \frac{m_n-1}{n}$$

$$\Rightarrow P \geq \lim_{n \rightarrow \infty} \frac{m_n-1}{n} = S$$

That is every upper bound is larger than S .